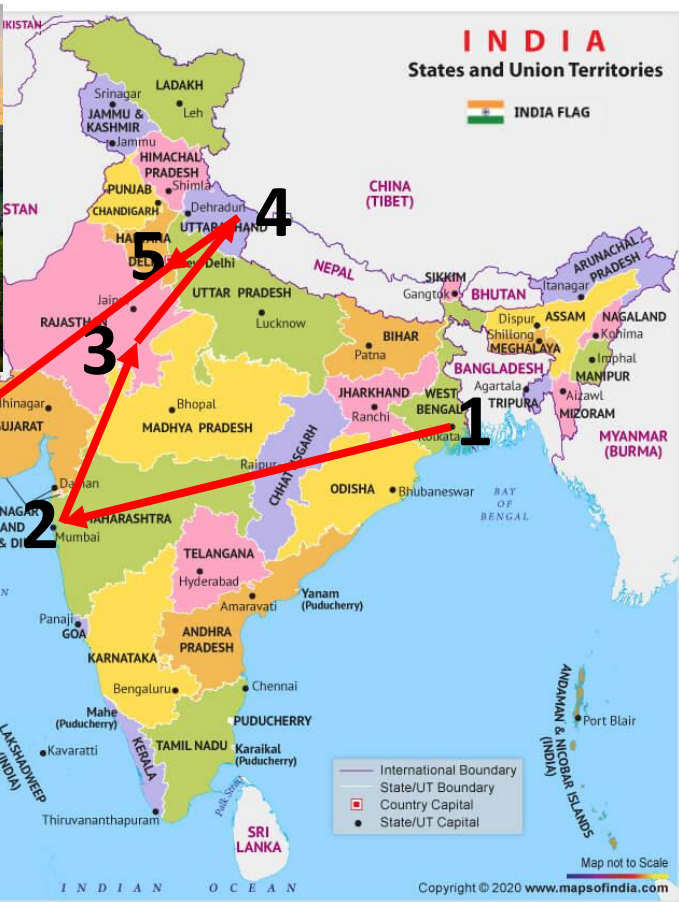


Scenario-based Optimization Models for Power Grid Resilience to Extreme Flooding Events

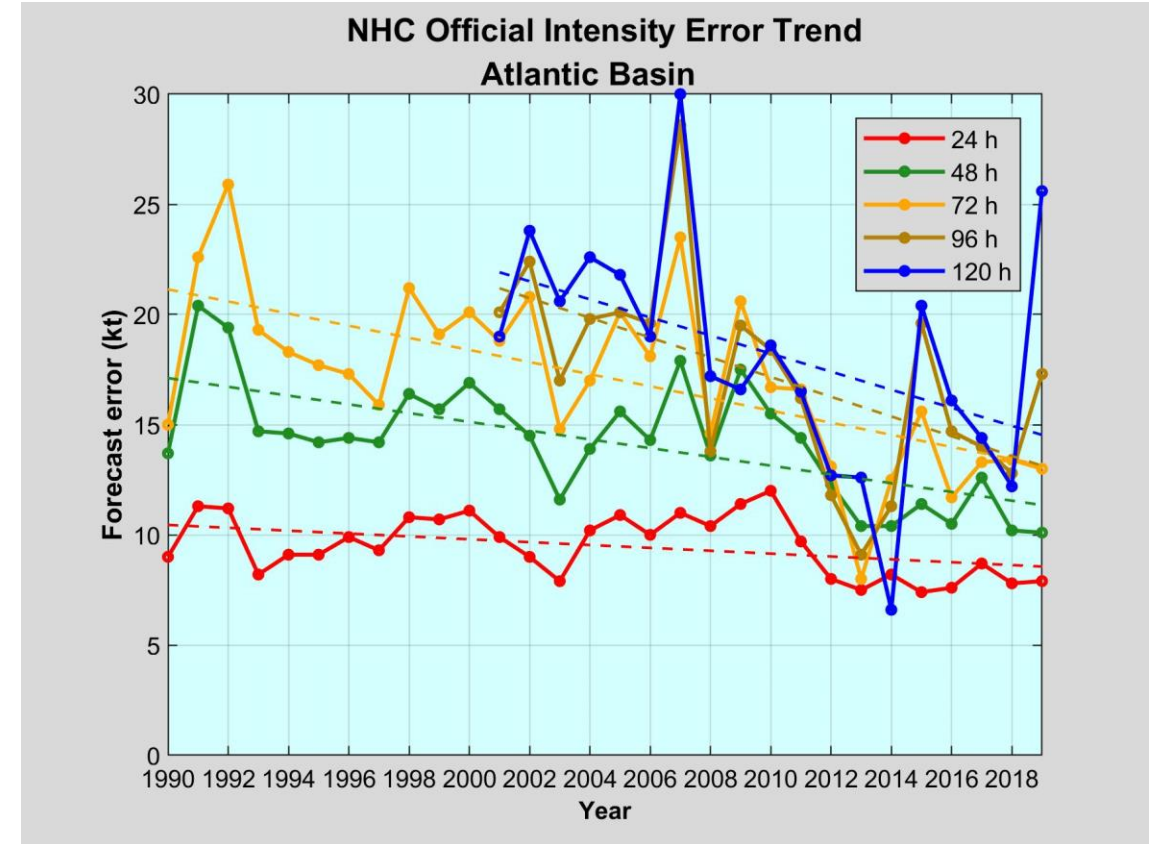
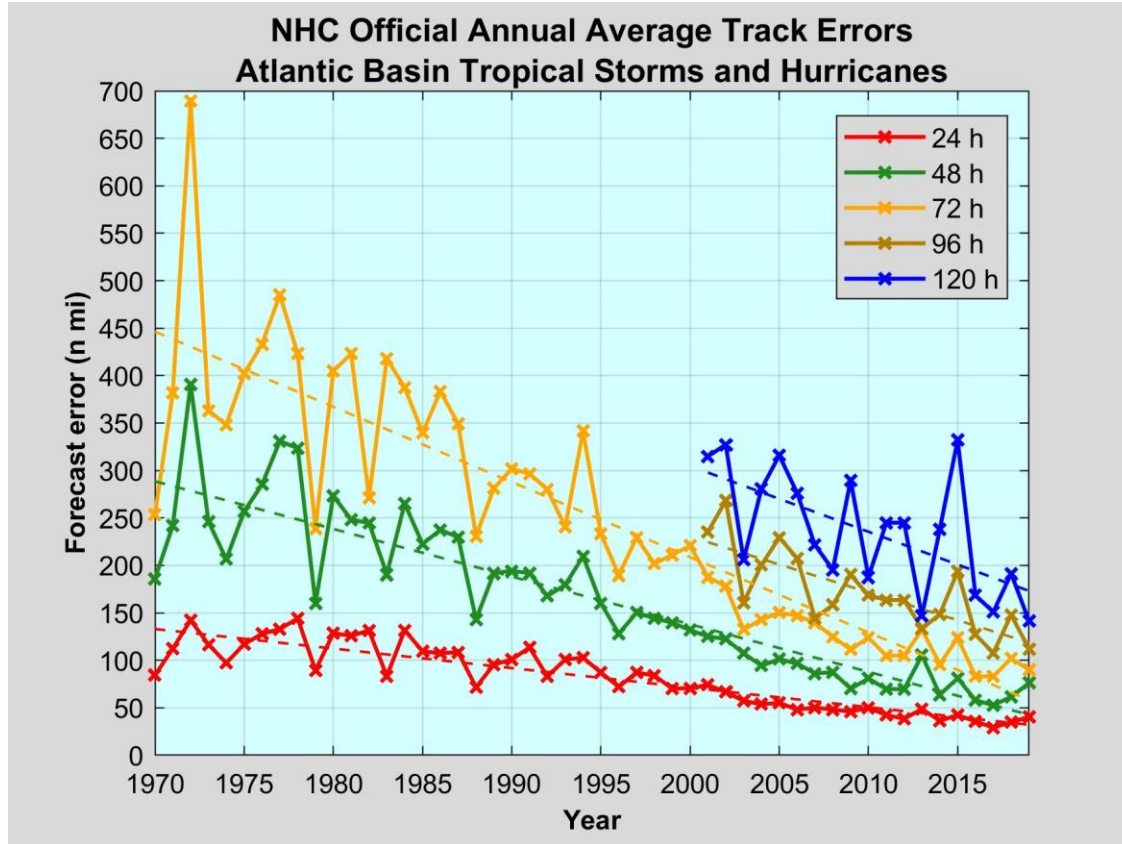
Ashutosh Shukla

PhD Candidate, Graduate program in ORIE, UT Austin

Dr. Erhan Kutanoglu and Dr. John J. Hasenbein



Weather forecasting has improved

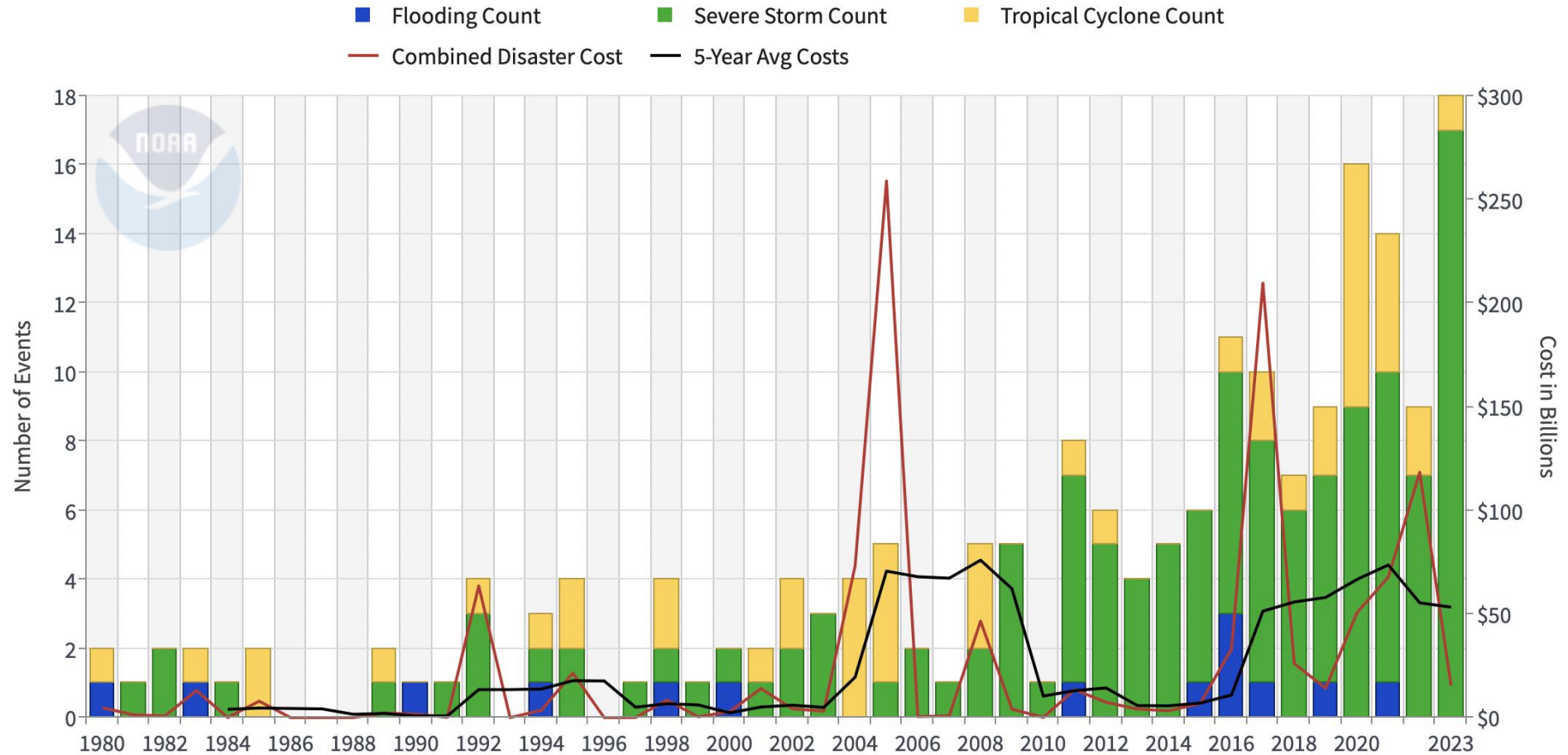


NOAA

Disaster costs continue to rise

Gulf Coast States Billion-Dollar Disaster Events 1980-2023 (CPI-Adjusted)

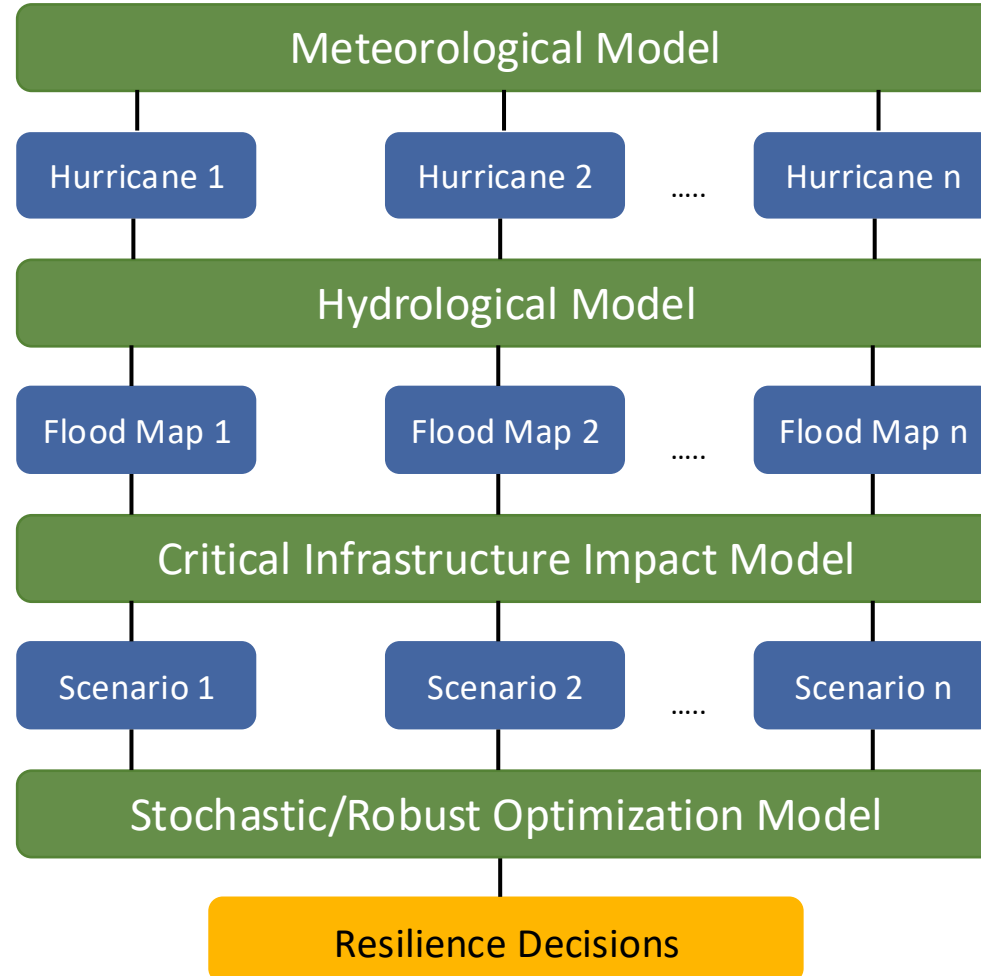
AL, FL, LA, MS, TX



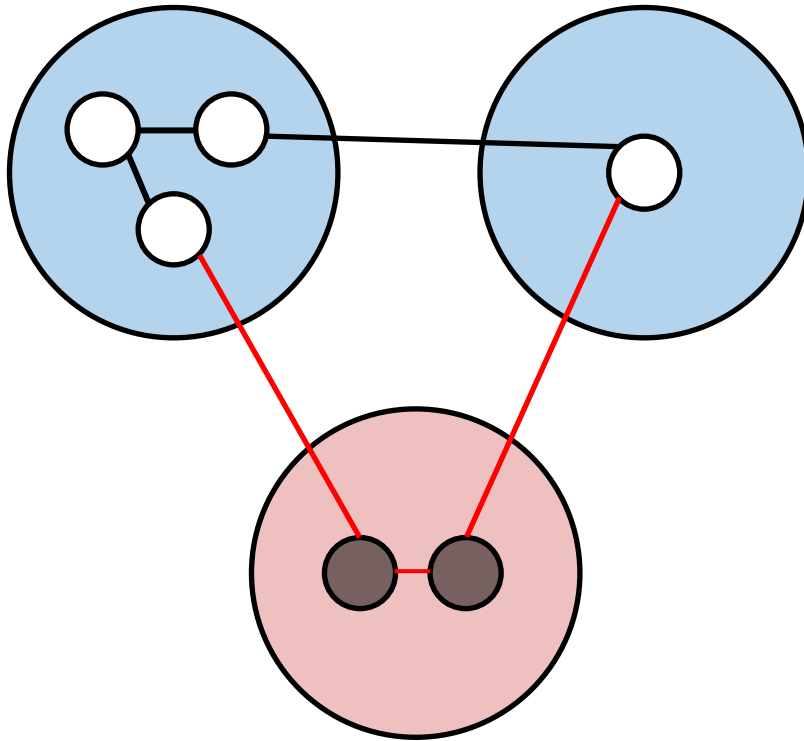
Updated: November 8, 2023

Powered by ZingChart

Proposed integrated decision-making framework



Transmission Grid Resilience to Flooding



Source: CenterPoint Energy's Flooded Substation in Houston during Hurricane Harvey

Mitigation and response measures



Source: <https://www.permacastwalls.com/gallery-2>

Two-Stage Stochastic Optimization Model

$$L_{SO}^* = \min_{x \in \mathcal{X}} \sum_{k \in \mathcal{K}} p_k \mathcal{L}(x, k)$$

$$\mathcal{X} = \left\{ \sum_{i \in \mathcal{I}_f} f_i y_i + v_i x_i \leq I, x_i \leq H_i y_i, \forall i \in \mathcal{I}_f \right\}$$

- Compact representation of the two-stage stochastic programming model
- Feasible set for the first-stage decisions



Two-Stage Stochastic Optimization Model

$$L_{SO}^* = \min_{x \in \mathcal{X}} \sum_{k \in \mathcal{K}} p_k \mathcal{L}(x, k)$$

$$\mathcal{L}(x, k) = \text{minimize} \quad \sum_{j \in \mathcal{J}} D_j - s_j,$$

subject to

$$\begin{aligned} (1 - z_j)M &\geq \Delta_{\theta(j)k} - x_{\theta(j)}, & \forall j : \theta(j) \in \mathcal{I}_f, \\ 2z_jM &\geq 1 - 2(\Delta_{\theta(j)k} - x_{\theta(j)}), & \forall j : \theta(j) \in \mathcal{I}_f, \\ z_j &= 1, & \forall j : \theta(j) \in \mathcal{I} \setminus \mathcal{I}_f, \end{aligned}$$

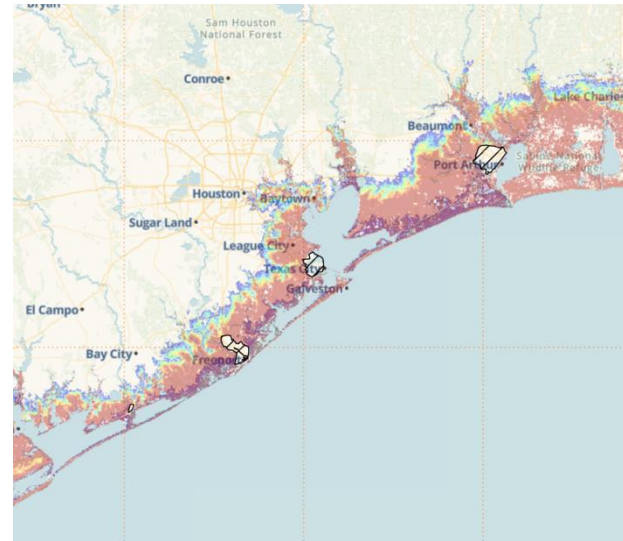
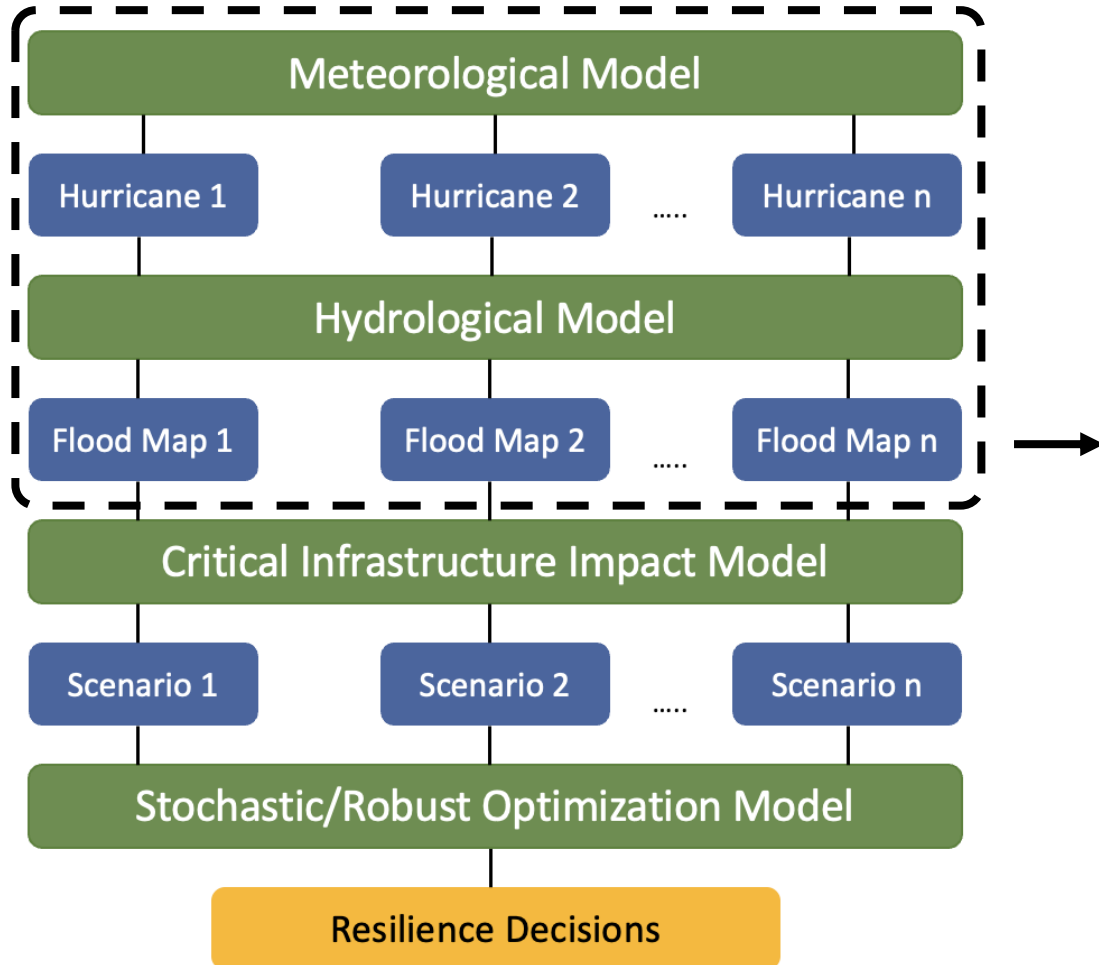
$$\begin{aligned} u_j &\leq z_j, & \forall j \in \mathcal{J}, \\ s_j &\leq z_j D_j, & \forall j \in \mathcal{J}, \\ u_j \underline{G}_j &\leq g_j \leq u_j \bar{G}_j, & \forall j \in \mathcal{J}, \\ -z_{\lambda(r)} F_r &\leq e_r \leq z_{\lambda(r)} F_r, & \forall r \in \mathcal{R}, \\ -z_{\mu(r)} F_r &\leq e_r \leq z_{\mu(r)} F_r, & \forall r \in \mathcal{R}, \\ B_r(\alpha_{\lambda(r)} - \alpha_{\mu(r)}) &\geq M(z_{\lambda(r)} + z_{\mu(r)}) - 2M + e_r, & \forall r \in \mathcal{R}, \\ B_r(\alpha_{\lambda(r)} - \alpha_{\mu(r)}) &\leq -M(z_{\lambda(r)} + z_{\mu(r)}) + 2M + e_r, & \forall r \in \mathcal{R}, \\ \sum_{r \in \mathcal{N}_j^{\text{out}}} e_r - \sum_{r \in \mathcal{N}_j^{\text{in}}} e_r &= g_j - s_j, & \forall j \in \mathcal{J}, \\ -\pi &\leq \alpha_j \leq \pi, & \forall j \in \mathcal{J}, \\ \alpha_\beta &= 0. \end{aligned}$$

Load shed

Linking Constraints

DC-OPF Constraints

Case Study: Correlated Flooding

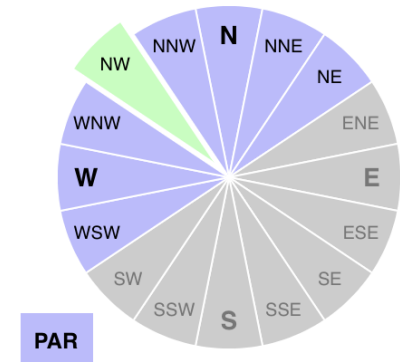


Category:



Additional Filters (MEOWs)

Direction:

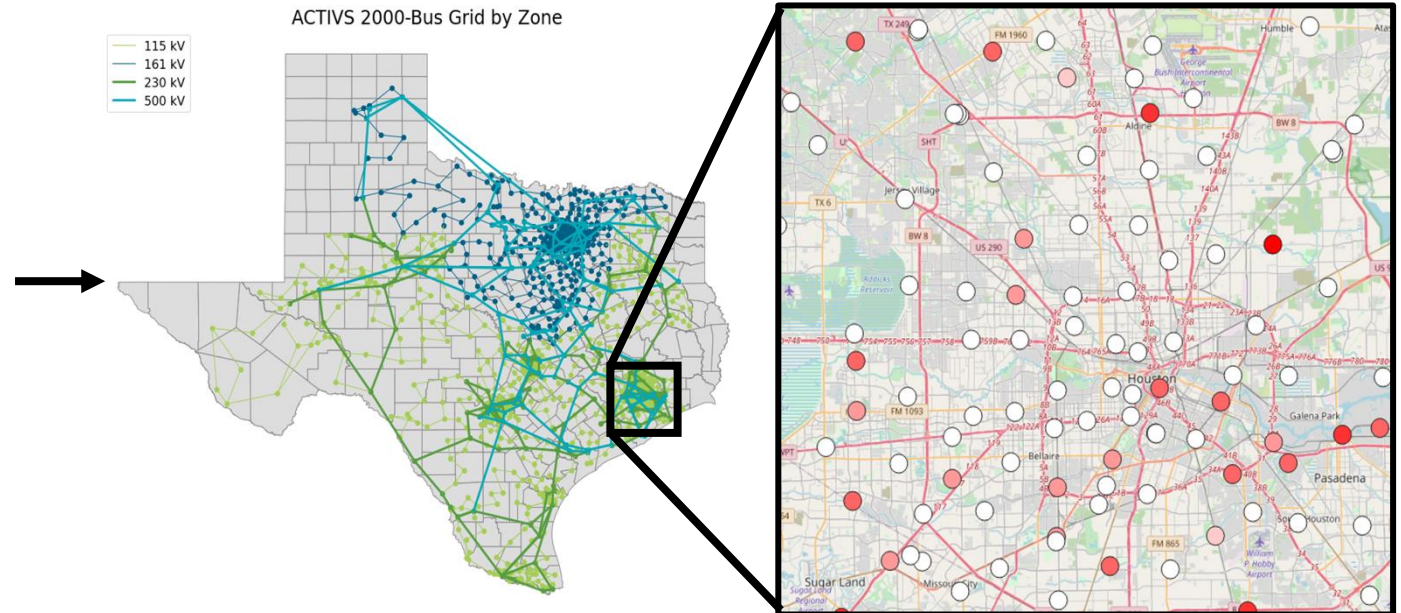
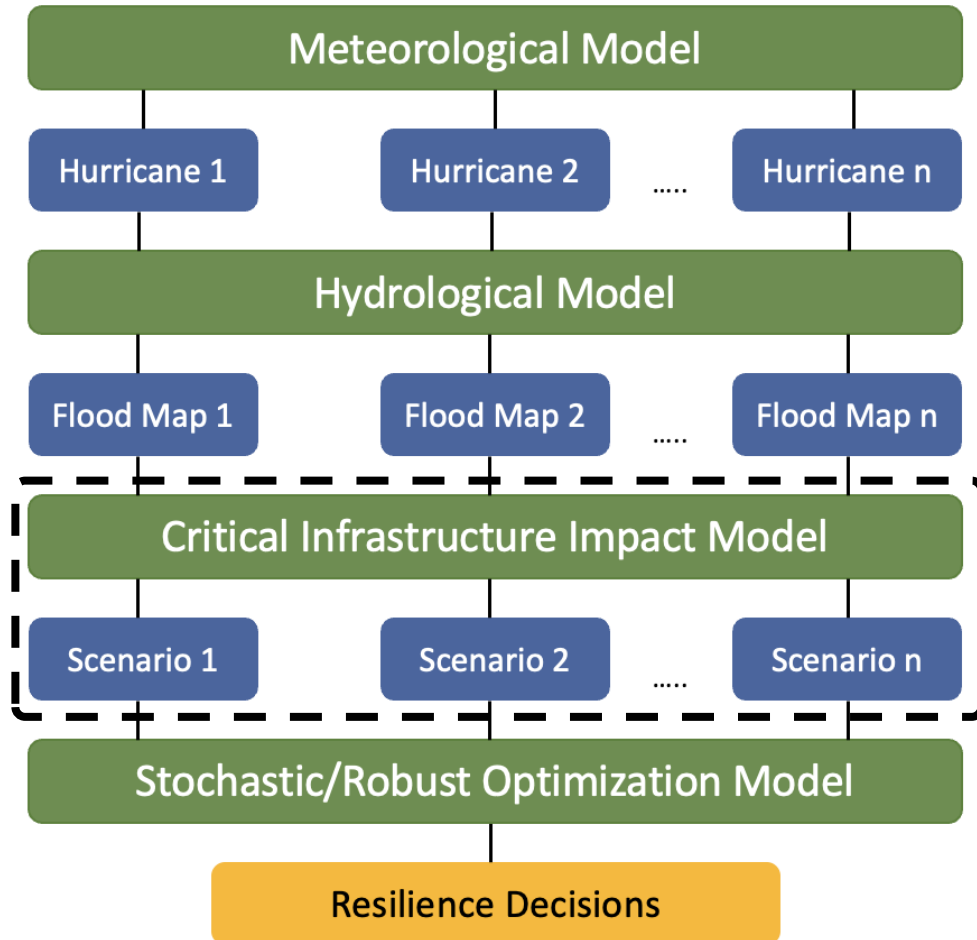


Forward Speed:



Storm-surge maps developed by NOAA

Case Study: Scenario Generation

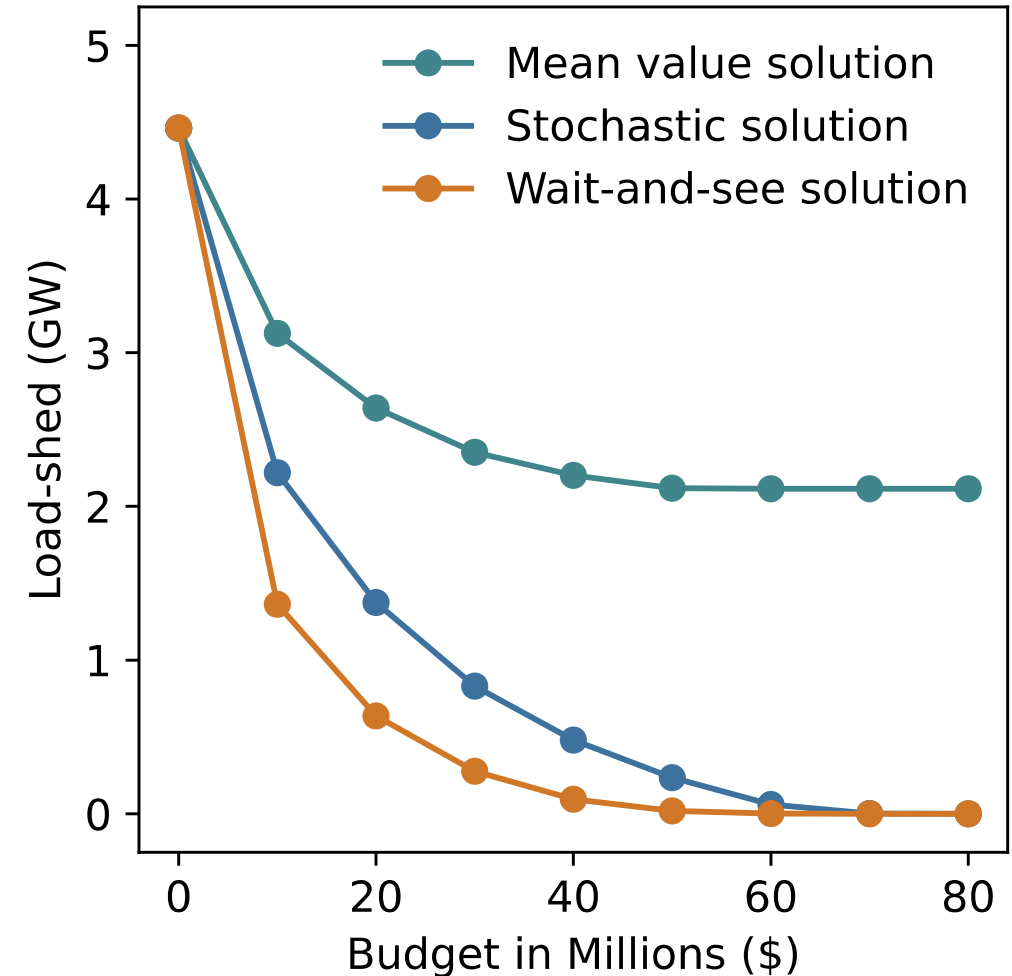


Stochastic Solutions and Bounds

$$L_{SO}^* = \min_{x \in \mathcal{X}} \sum_{k \in \mathcal{K}} p_k \mathcal{L}(x, k)$$

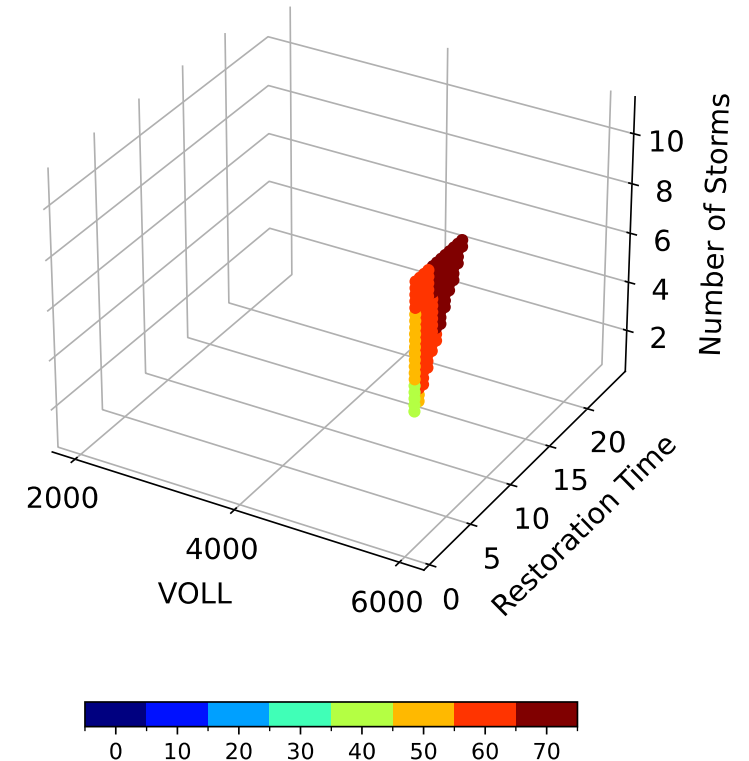
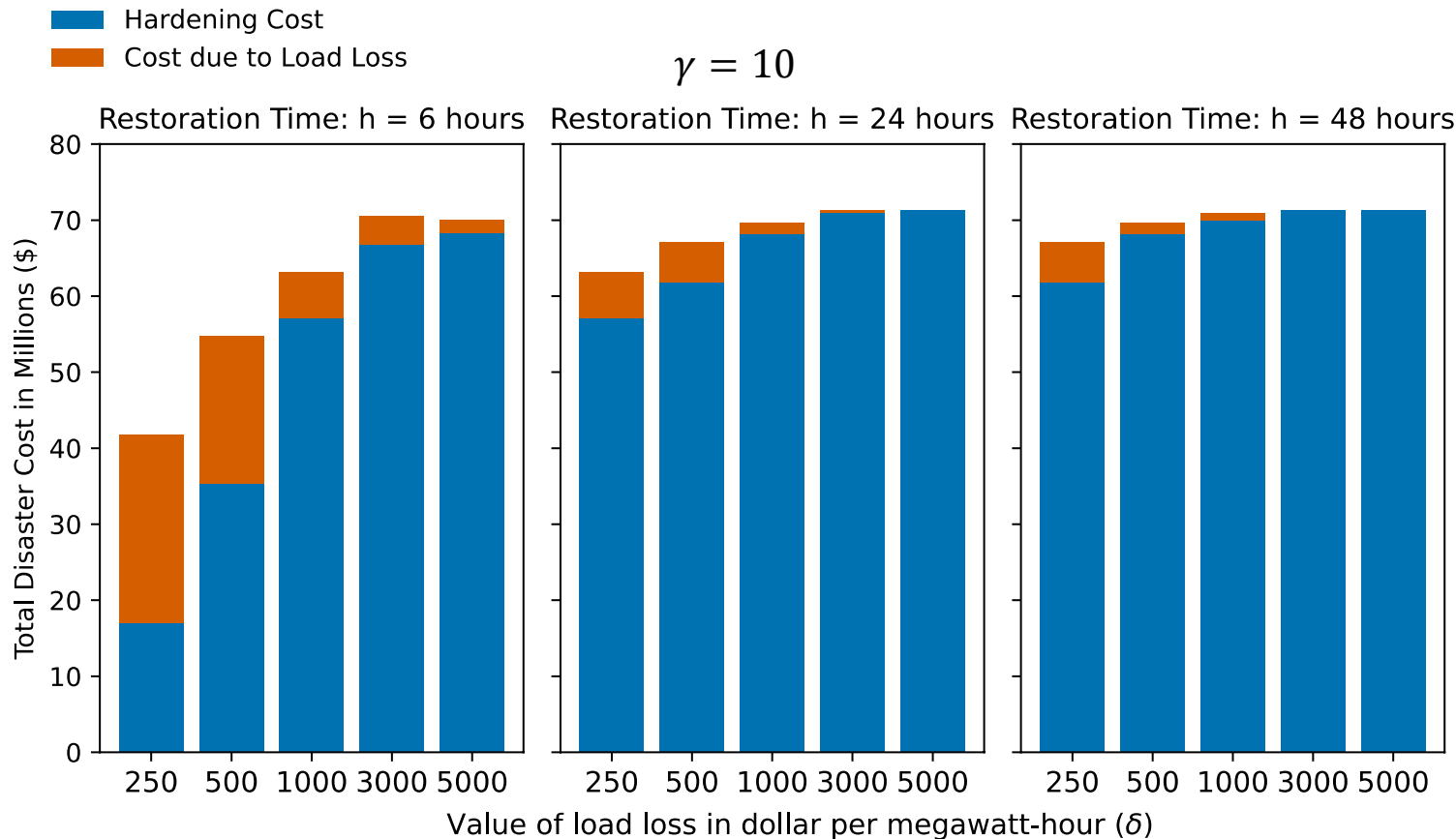
$$\mathcal{X} = \left\{ \sum_{i \in \mathcal{I}_f} f_i y_i + v_i x_i \leq I, x_i \leq H_i y_i, \forall i \in \mathcal{I}_f \right\}$$

- Compact representation of the two-stage stochastic programming model
- Feasible set for the first-stage decisions



Determining optimal budget for substation hardening

$$TDM_{SO} = \left\{ \min_{\gamma h \delta} \left(\sum_{k \in \mathcal{K}} p_k \mathcal{L}(x, k) \right) + \sum_{i \in \mathcal{I}_f} f_i y_i + v_i x_i : x_i \leq H_i y_i, \forall i \in \mathcal{I}_f \right\}$$



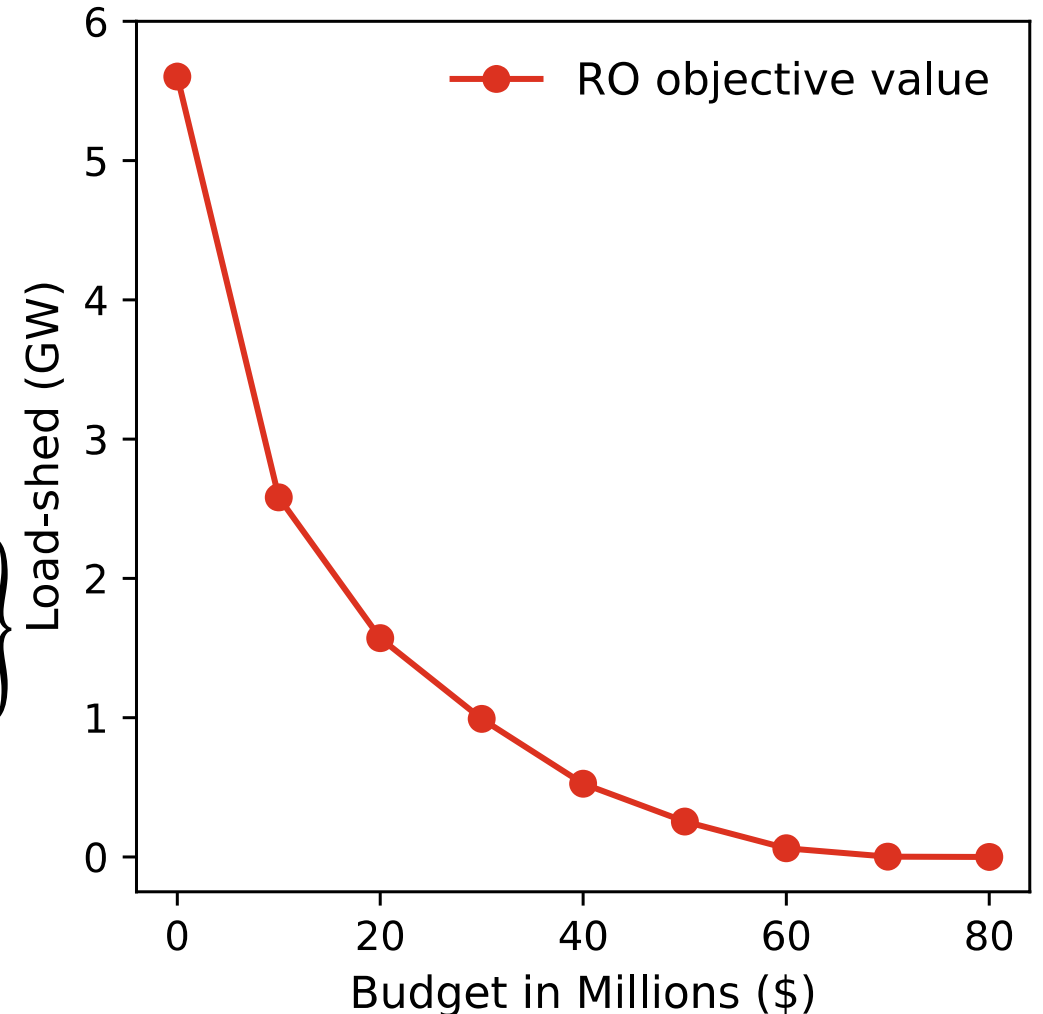
Two-Stage Robust Optimization Model

$$L_{SO}^* = \min_{x \in \mathcal{X}} \left(\sum_{k \in \mathcal{K}} p_k \mathcal{L}(x, k) \right)$$

↓

$$L_{RO}^* = \min_{x \in \mathcal{X}} \left(\max_{k \in \mathcal{K}} \mathcal{L}(x, k) \right)$$

$$\mathcal{X} = \left\{ \sum_{i \in \mathcal{I}_f} f_i y_i + v_i x_i \leq I, x_i \leq H_i y_i, \forall i \in \mathcal{I}_f \right\}$$



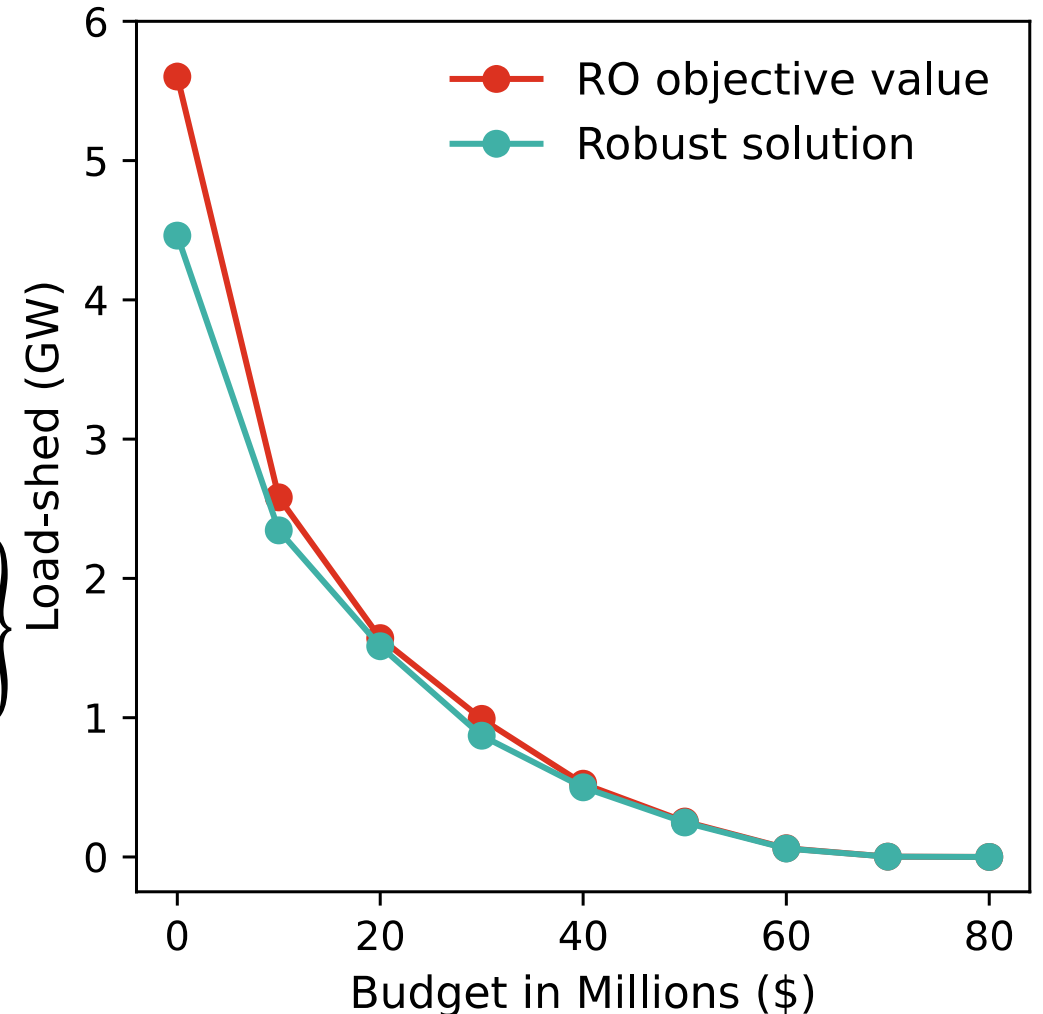
Two-Stage Robust Optimization Model

$$L_{SO}^* = \min_{x \in \mathcal{X}} \left(\sum_{k \in \mathcal{K}} p_k \mathcal{L}(x, k) \right)$$

↓

$$L_{RO}^* = \min_{x \in \mathcal{X}} \left(\max_{k \in \mathcal{K}} \mathcal{L}(x, k) \right)$$

$$\mathcal{X} = \left\{ \sum_{i \in \mathcal{I}_f} f_i y_i + v_i x_i \leq I, x_i \leq H_i y_i, \forall i \in \mathcal{I}_f \right\}$$



Two-Stage Robust Optimization Model

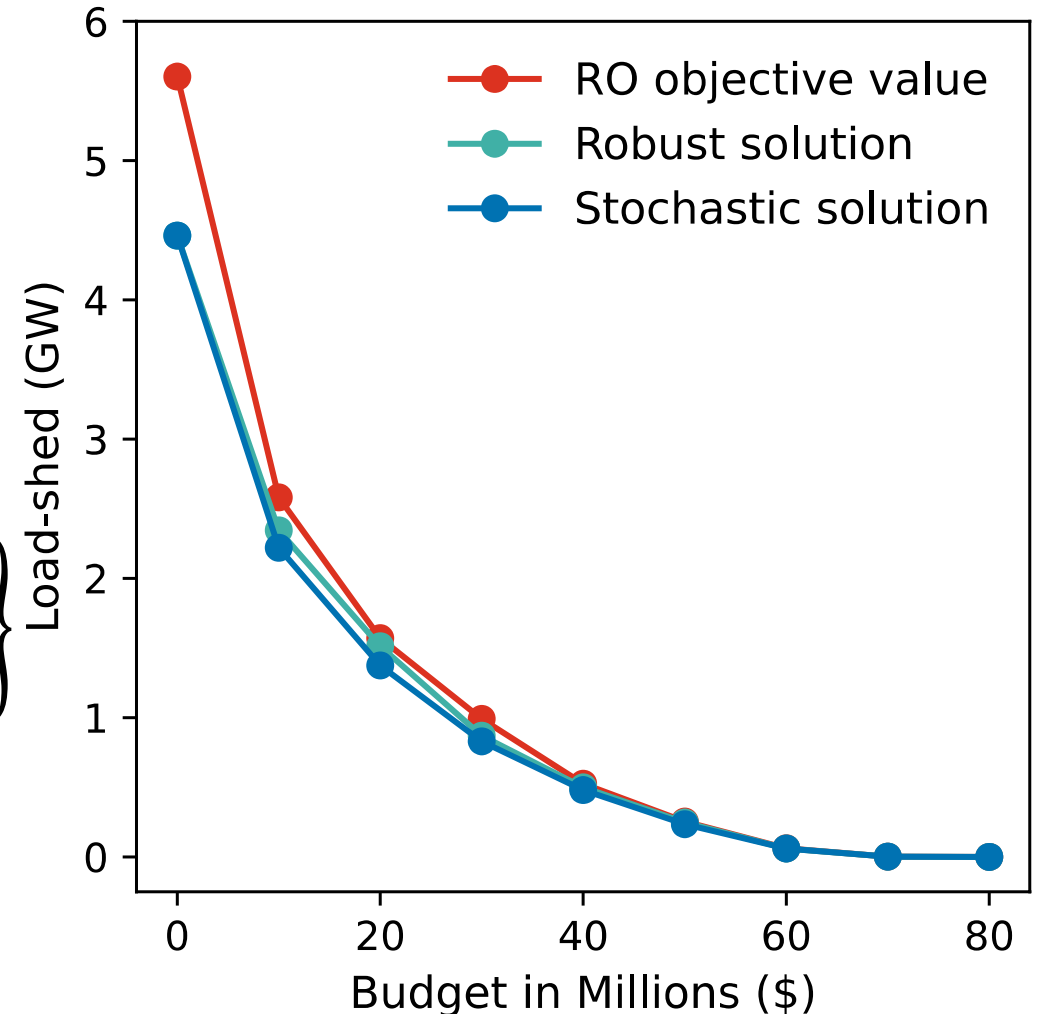
$$L_{SO}^* = \min_{x \in \mathcal{X}} \left(\sum_{k \in \mathcal{K}} p_k \mathcal{L}(x, k) \right)$$

↓

$$L_{RO}^* = \min_{x \in \mathcal{X}} \left(\max_{k \in \mathcal{K}} \mathcal{L}(x, k) \right)$$

$$\mathcal{X} = \left\{ \sum_{i \in \mathcal{I}_f} f_i y_i + v_i x_i \leq I, x_i \leq H_i y_i, \forall i \in \mathcal{I}_f \right\}$$

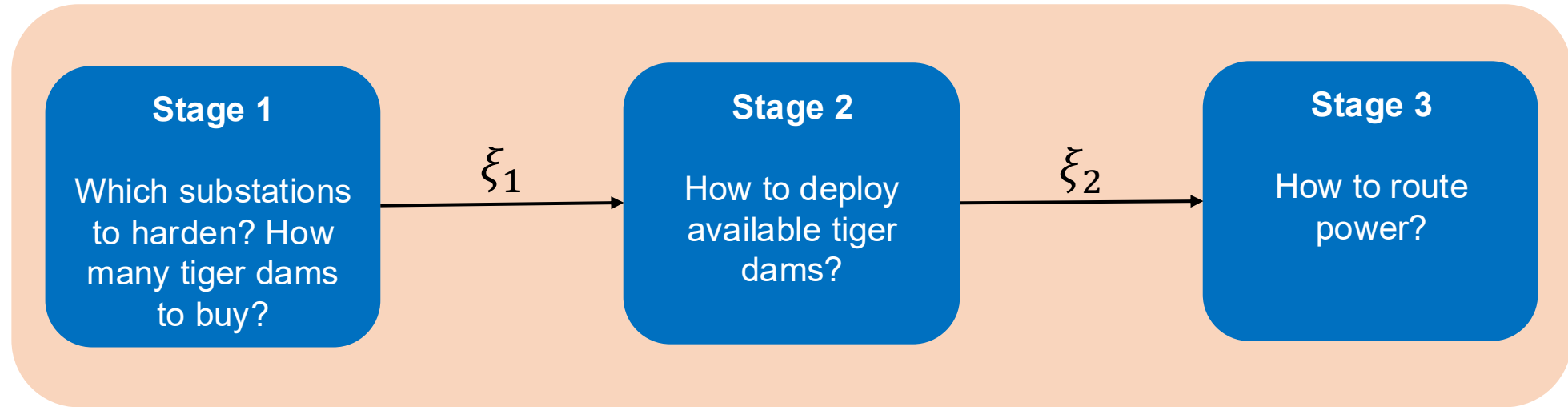
Proposition: The optimal objective value of the stochastic program is upper bounded by the objective function value of the robust optimization problem irrespective of the probabilities associated with the scenarios.



Budget Allocation in Optimizing Power Grid Resilience to Extreme Flooding Events: Hardening, Preparation, or Recovery



Proposed Model Overview: Decision-making Timeline



Mathematical Formulation

$$\begin{aligned}
 \min \quad & \underbrace{\sum_{i \in \mathcal{I}} (C_f y_i + C_v x_i) + n C_t + TC_o \sum_{k \in \mathcal{P}} \sum_{i \in \mathcal{I}} \text{Pr}(k) q_{ik} + TV \sum_{k \in \mathcal{P}} \sum_{m \in \mathcal{Q}(k)} \text{Pr}(k, m) l_{km}}_{\text{Expected cost due to flooding}} \quad (1a) \\
 \text{s.t.} \quad & x_i \leq W_i y_i \quad \forall i \in \mathcal{I} \quad (1b) \\
 & p_{ik} \leq H q_{ik} \quad \forall i \in \mathcal{I}, k \in \mathcal{P} \quad (1c) \\
 & \sum_{i \in \mathcal{I}} p_{ik} \leq n, \quad \forall k \in \mathcal{P} \quad (1d) \\
 & \underbrace{z_{jkm} \zeta \geq 2 \max(p_{\theta(j)k}, x_{\theta(j)}) - 2\Delta_{jkm} + 1}_{\text{Cost of Substation Hardening}} \quad (1e) \\
 & u_{jkm} \leq z_{jkm}, \quad \forall j \in \mathcal{J}, k \in \mathcal{P}, m \in \mathcal{Q}(k) \quad (1f) \\
 & s_{jkm} \leq z_{jkm} D_j, \quad \forall j \in \mathcal{J}, k \in \mathcal{P}, m \in \mathcal{Q}(k) \quad (1g) \\
 & \underbrace{u_{jkm} \underline{G}_j \leq g_{jkm} \leq u_{jkm} \overline{G}_j}_{\text{Acquisition Cost of Tiger Dams}} \quad (1h) \\
 & -z_{\lambda(r)km} F_r \leq e_{rkm} \leq z_{\lambda(r)km} F_r, \quad \forall r \in \mathcal{R}, k \in \mathcal{P}, m \in \mathcal{Q}(k) \quad (1i) \\
 & -z_{\mu(r)km} F_r \leq e_{rkm} \leq z_{\mu(r)km} F_r, \quad \forall r \in \mathcal{R}, k \in \mathcal{P}, m \in \mathcal{Q}(k) \quad (1j) \\
 & B_r(v_{\lambda(r)km} - v_{\mu(r)km}) \geq M(z_{\lambda(r)km} + z_{\mu(r)km}) - 2M + e_{rkm}, \quad \forall r \in \mathcal{R}, k \in \mathcal{P}, m \in \mathcal{Q}(k) \quad (1k) \\
 & B_r(v_{\lambda(r)km} - v_{\mu(r)km}) \leq -M(z_{\lambda(r)km} + z_{\mu(r)km}) + 2M + e_{rkm}, \quad \forall r \in \mathcal{R}, k \in \mathcal{P}, m \in \mathcal{Q}(k) \quad (1l) \\
 & \sum_{r \in \mathcal{N}_j} e_{rkm} = g_{jkm} - s_{jkm}, \quad \forall j \in \mathcal{J}, k \in \mathcal{P}, m \in \mathcal{Q}(k) \quad (1m) \\
 & v_{\beta km} = 0, \quad \forall k \in \mathcal{P}, m \in \mathcal{Q}(k) \quad (1n) \\
 & \sum_{j \in \mathcal{J}} D_j - s_{jkm} = l_{km}, \quad \forall k \in \mathcal{P}, m \in \mathcal{Q}(k) \quad (1p)
 \end{aligned}$$

Expected VOLL over T Hurricanes

Mathematical Formulation

$$\begin{aligned}
 \min \quad & \boxed{\sum_{i \in \mathcal{I}} (C_f y_i + C_v x_i)} + \boxed{n C_t} + \boxed{TC_o \sum_{k \in \mathcal{P}} \sum_{i \in \mathcal{I}} \text{Pr}(k) q_{ik}} + \boxed{TV \sum_{k \in \mathcal{P}} \sum_{m \in \mathcal{Q}(k)} \text{Pr}(k, m) l_{km}}, & (1a) \\
 \text{s.t.} \quad & x_i \leq W_i y_i, & \forall i \in \mathcal{I} & (1b) \\
 & p_{ik} \leq H q_{ik}, & \forall i \in \mathcal{I}, k \in \mathcal{P} & (1c) \\
 & \sum_{i \in \mathcal{I}} p_{ik} \leq n, & \forall k \in \mathcal{P} & (1d) \\
 & (1 - z_{jkm}) \zeta \geq \Delta_{jkm} - \max(x_{\theta(j)}, p_{\theta(j)k}), & \forall j \in \mathcal{J}, k \in \mathcal{P}, m \in \mathcal{Q}(k) & (1e) \\
 & 2z_{jkm} \zeta \geq 2 \max(p_{\theta(j)k}, x_{\theta(j)}) - 2\Delta_{jkm} + 1, & \forall j \in \mathcal{J}, k \in \mathcal{P}, m \in \mathcal{Q}(k) & (1f) \\
 & u_{jkm} \leq z_{jkm}, & \forall j \in \mathcal{J}, k \in \mathcal{P}, m \in \mathcal{Q}(k) & (1g) \\
 & s_{jkm} \leq z_{jkm} D_j, & \forall j \in \mathcal{J}, k \in \mathcal{P}, m \in \mathcal{Q}(k) & (1h) \\
 & u_{jkm} \underline{G}_j \leq g_{jkm} \leq u_{jkm} \overline{G}_j, & j \in \mathcal{J}, k \in \mathcal{P}, m \in \mathcal{Q}(k) & (1i) \\
 & -z_{\lambda(r)km} F_r \leq e_{rkm} \leq z_{\lambda(r)km} F_r, & \forall r \in \mathcal{R}, k \in \mathcal{P}, m \in \mathcal{Q}(k) & (1j) \\
 & -z_{\mu(r)km} F_r \leq e_{rkm} \leq z_{\mu(r)km} F_r, & \forall r \in \mathcal{R}, k \in \mathcal{P}, m \in \mathcal{Q}(k) & (1k) \\
 & B_r(v_{\lambda(r)km} - v_{\mu(r)km}) \geq M(z_{\lambda(r)km} + z_{\mu(r)km}) - 2M + e_{rkm}, & \forall r \in \mathcal{R}, k \in \mathcal{P}, m \in \mathcal{Q}(k) & (1l) \\
 & B_r(v_{\lambda(r)km} - v_{\mu(r)km}) \leq -M(z_{\lambda(r)km} + z_{\mu(r)km}) + 2M + e_{rkm}, & \forall r \in \mathcal{R}, k \in \mathcal{P}, m \in \mathcal{Q}(k) & (1m) \\
 & \sum_{r \in \mathcal{N}_j} e_{rkm} = g_{jkm} - s_{jkm}, & \forall j \in \mathcal{J}, k \in \mathcal{P}, m \in \mathcal{Q}(k) & (1n) \\
 & v_{\beta km} = 0, & \forall k \in \mathcal{P}, m \in \mathcal{Q}(k) & (1o) \\
 & \sum_{j \in \mathcal{J}} D_j - s_{jkm} = l_{km}. & \forall k \in \mathcal{P}, m \in \mathcal{Q}(k) & (1p)
 \end{aligned}$$

Expected cost due to flooding

Technological Constraints

Mathematical Formulation

$$\begin{aligned}
 \min \quad & \boxed{\sum_{i \in \mathcal{I}} (C_f y_i + C_v x_i)} + \boxed{n C_t} + \boxed{TC_o \sum_{k \in \mathcal{P}} \sum_{i \in \mathcal{I}} \text{Pr}(k) q_{ik}} + \boxed{TV \sum_{k \in \mathcal{P}} \sum_{m \in \mathcal{Q}(k)} \text{Pr}(k, m) l_{km}}, & (1a) \\
 \text{s.t.} \quad & x_i \leq W_i y_i, & \forall i \in \mathcal{I} & (1b) \\
 & p_{ik} \leq H q_{ik}, & \forall i \in \mathcal{I}, k \in \mathcal{P} & (1c) \\
 & \sum_{i \in \mathcal{I}} p_{ik} \leq n, & \forall k \in \mathcal{P} & (1d) \\
 & (1 - z_{jkm}) \zeta \geq \Delta_{jkm} - \max(x_{\theta(j)}, p_{\theta(j)k}), & \forall j \in \mathcal{J}, k \in \mathcal{P}, m \in \mathcal{Q}(k) & (1e) \\
 & 2z_{jkm} \zeta \geq 2 \max(p_{\theta(j)k}, x_{\theta(j)}) - 2\Delta_{jkm} + 1, & \forall j \in \mathcal{J}, k \in \mathcal{P}, m \in \mathcal{Q}(k) & (1f) \\
 & u_{jkm} \leq z_{jkm}, & \forall j \in \mathcal{J}, k \in \mathcal{P}, m \in \mathcal{Q}(k) & (1g) \\
 & s_{jkm} \leq z_{jkm} D_j, & \forall j \in \mathcal{J}, k \in \mathcal{P}, m \in \mathcal{Q}(k) & (1h) \\
 & u_{jkm} \underline{G}_j \leq g_{jkm} \leq u_{jkm} \overline{G}_j, & j \in \mathcal{J}, k \in \mathcal{P}, m \in \mathcal{Q}(k) & (1i) \\
 & -z_{\lambda(r)km} F_r \leq e_{rkm} \leq z_{\lambda(r)km} F_r, & \forall r \in \mathcal{R}, k \in \mathcal{P}, m \in \mathcal{Q}(k) & (1j) \\
 & -z_{\mu(r)km} F_r \leq e_{rkm} \leq z_{\mu(r)km} F_r, & \forall r \in \mathcal{R}, k \in \mathcal{P}, m \in \mathcal{Q}(k) & (1k) \\
 & B_r(v_{\lambda(r)km} - v_{\mu(r)km}) \geq M(z_{\lambda(r)km} + z_{\mu(r)km}) - 2M + e_{rkm}, & \forall r \in \mathcal{R}, k \in \mathcal{P}, m \in \mathcal{Q}(k) & (1l) \\
 & B_r(v_{\lambda(r)km} - v_{\mu(r)km}) \leq -M(z_{\lambda(r)km} + z_{\mu(r)km}) + 2M + e_{rkm}, & \forall r \in \mathcal{R}, k \in \mathcal{P}, m \in \mathcal{Q}(k) & (1m) \\
 & \sum_{r \in \mathcal{N}_j} e_{rkm} = g_{jkm} - s_{jkm}, & \forall j \in \mathcal{J}, k \in \mathcal{P}, m \in \mathcal{Q}(k) & (1n) \\
 & v_{\beta km} = 0, & \forall k \in \mathcal{P}, m \in \mathcal{Q}(k) & (1o) \\
 & \sum_{j \in \mathcal{J}} D_j - s_{jkm} = l_{km}. & \forall k \in \mathcal{P}, m \in \mathcal{Q}(k) & (1p)
 \end{aligned}$$

Expected cost due to flooding

Technological Constraints

Linking Constraints

Mathematical Formulation

$$\begin{aligned}
 \min \quad & \boxed{\sum_{i \in \mathcal{I}} (C_f y_i + C_v x_i)} + \boxed{n C_t} + \boxed{TC_o \sum_{k \in \mathcal{P}} \sum_{i \in \mathcal{I}} \text{Pr}(k) q_{ik}} + \boxed{TV \sum_{k \in \mathcal{P}} \sum_{m \in \mathcal{Q}(k)} \text{Pr}(k, m) l_{km}}, & (1a) \\
 \text{s.t.} \quad & x_i \leq W_i y_i, & \forall i \in \mathcal{I} & (1b) \\
 & p_{ik} \leq H q_{ik}, & \forall i \in \mathcal{I}, k \in \mathcal{P} & (1c) \\
 & \sum_{i \in \mathcal{I}} p_{ik} \leq n, & \forall k \in \mathcal{P} & (1d) \\
 & (1 - z_{jkm}) \zeta \geq \Delta_{jkm} - \max(x_{\theta(j)}, p_{\theta(j)k}), & \forall j \in \mathcal{J}, k \in \mathcal{P}, m \in \mathcal{Q}(k) & (1e) \\
 & 2z_{jkm} \zeta \geq 2 \max(p_{\theta(j)k}, x_{\theta(j)}) - 2\Delta_{jkm} + 1, & \forall j \in \mathcal{J}, k \in \mathcal{P}, m \in \mathcal{Q}(k) & (1f) \\
 & u_{jkm} \leq z_{jkm}, & \forall j \in \mathcal{J}, k \in \mathcal{P}, m \in \mathcal{Q}(k) & (1g) \\
 & s_{jkm} \leq z_{jkm} D_j, & \forall j \in \mathcal{J}, k \in \mathcal{P}, m \in \mathcal{Q}(k) & (1h) \\
 & u_{jkm} \underline{G}_j \leq g_{jkm} \leq u_{jkm} \overline{G}_j, & j \in \mathcal{J}, k \in \mathcal{P}, m \in \mathcal{Q}(k) & (1i) \\
 & -z_{\lambda(r)km} F_r \leq e_{rkm} \leq z_{\lambda(r)km} F_r, & \forall r \in \mathcal{R}, k \in \mathcal{P}, m \in \mathcal{Q}(k) & (1j) \\
 & -z_{\mu(r)km} F_r \leq e_{rkm} \leq z_{\mu(r)km} F_r, & \forall r \in \mathcal{R}, k \in \mathcal{P}, m \in \mathcal{Q}(k) & (1k) \\
 & B_r(v_{\lambda(r)km} - v_{\mu(r)km}) \geq M(z_{\lambda(r)km} + z_{\mu(r)km}) - 2M + e_{rkm}, & \forall r \in \mathcal{R}, k \in \mathcal{P}, m \in \mathcal{Q}(k) & (1l) \\
 & B_r(v_{\lambda(r)km} - v_{\mu(r)km}) \leq -M(z_{\lambda(r)km} + z_{\mu(r)km}) + 2M + e_{rkm}, & \forall r \in \mathcal{R}, k \in \mathcal{P}, m \in \mathcal{Q}(k) & (1m) \\
 & \sum_{r \in \mathcal{N}_j} e_{rkm} = g_{jkm} - s_{jkm}, & \forall j \in \mathcal{J}, k \in \mathcal{P}, m \in \mathcal{Q}(k) & (1n) \\
 & v_{\beta km} = 0, & \forall k \in \mathcal{P}, m \in \mathcal{Q}(k) & (1o) \\
 & \sum_{j \in \mathcal{J}} D_j - s_{jkm} = l_{km}. & \forall k \in \mathcal{P}, m \in \mathcal{Q}(k) & (1p)
 \end{aligned}$$

Expected cost due to flooding

Technological Constraints

Linking Constraints

DC-OPF Constraints

Questions that we answer with the three-stage model

- How optimal budget allocation changes with
 - the value of load loss
 - the power restoration time in the aftermath of the hurricane
 - the improvement in design of temporary flood barriers
 - the change in the cost of substation hardening
- How to quantify the gains from coordinated decision making?
- How access to perfect forecasts impacts optimal budget allocation?

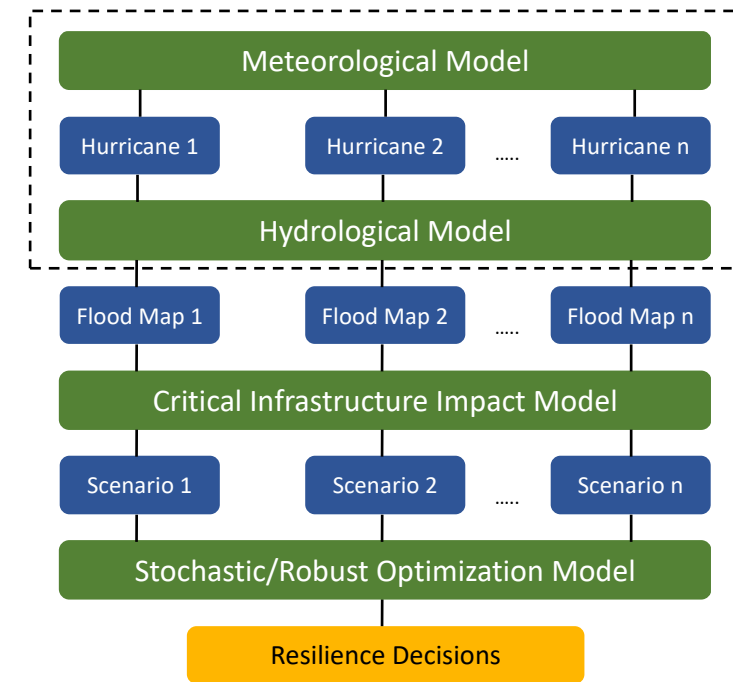
Ongoing and future research

Ongoing Research

- Input modeling for extreme flood scenario generation

Plans for future research

- Model other extreme weather threats like wildfire, extreme cold events
- Make methodological contributions to account for equity in resilience decision-making
- Develop optimization models for resilience of the distribution grid
- Develop models to capture the interdependence of critical infrastructure networks



Acknowledgements

- Professor Surya Santoso – Department of Electrical and Computer Engineering
- Professor Zong-Liang Yang – Jackson School of Geosciences
- Professor Carey King – Energy Institute
- Professor Clint Dawson – Oden Institute of Computational Sciences
- Brent Austgen – PhD Student, ORIE program
- Dr. Wen-Ying Wu – Jackson School of Geosciences
- Joshua Yip – PhD Student, Department of Electrical and Computer Engineering
- Sabiha Tabassum – PhD Student, Jackson School of Geosciences



The University of Texas at Austin
Energy Institute

OPERATIONS RESEARCH AND INDUSTRIAL ENGINEERING

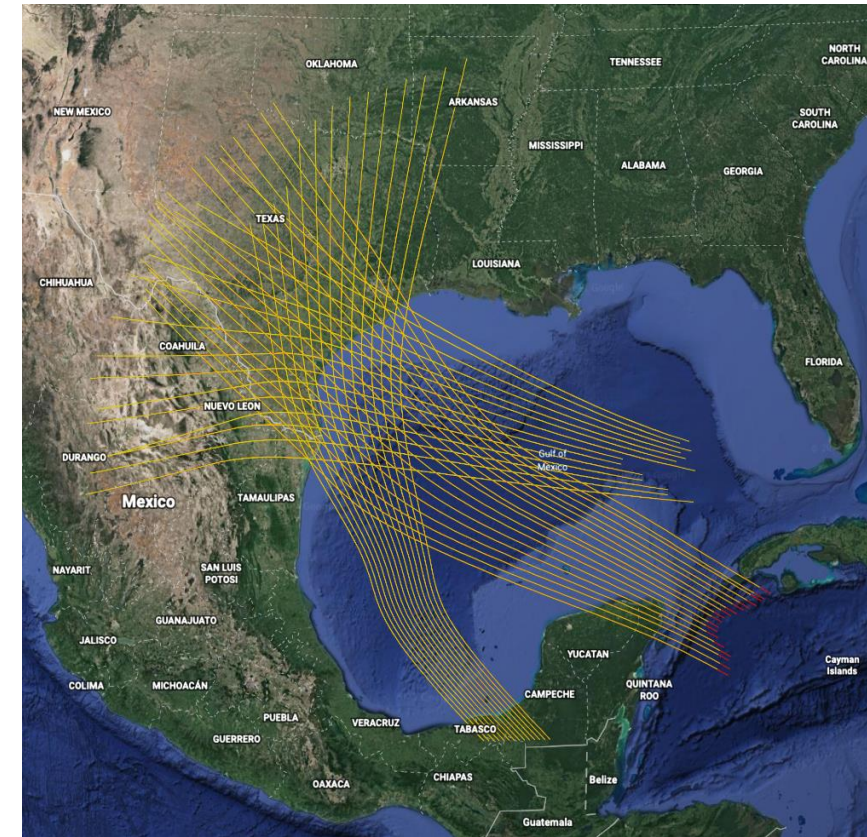
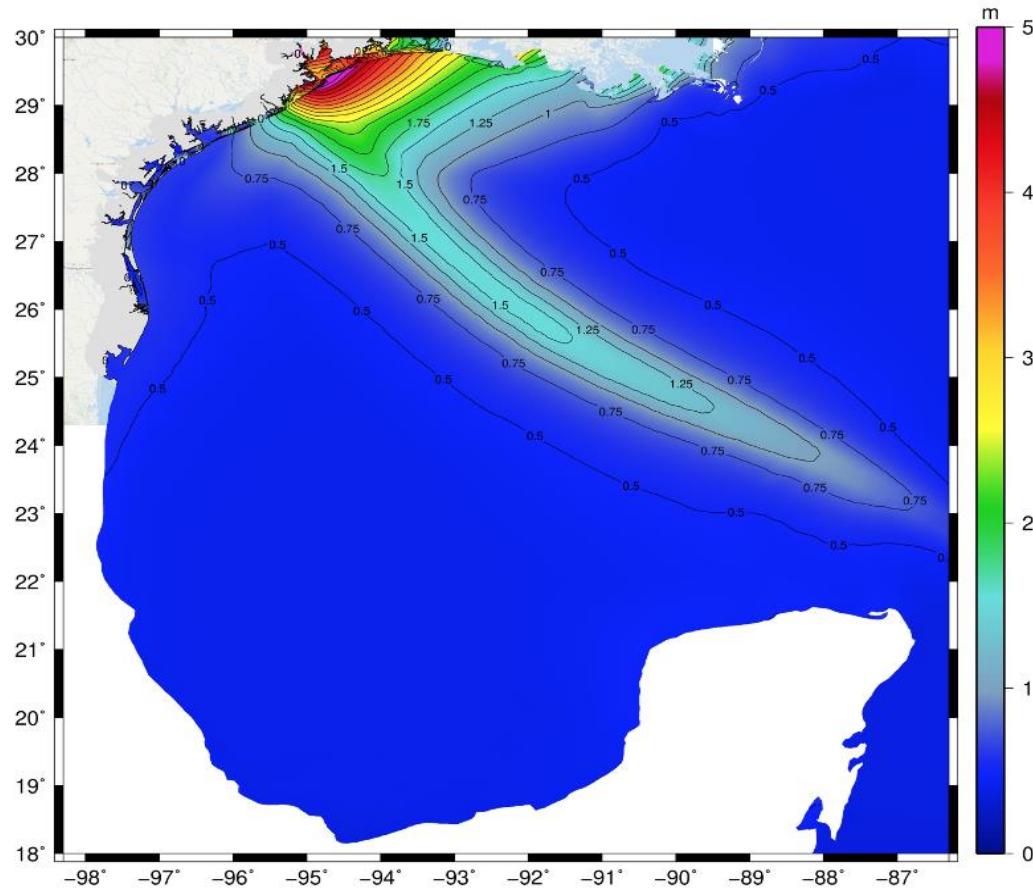
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Thank you

Appendix

Synthetic Storms using ADCIRC



Dawson, C. Del-Castillo-Negrete, C. **Shukla**, A. PACHEV, B. Kaiser, C. Kutanoglu, E. (2021) "ADCIRC Simulation of Synthetic Storms in the Gulf of Mexico", in *Texas FEMA Hurricane Winds and Surge*. DesignSafe-CI. <https://doi.org/10.17603/ds2-68a9-0s64>.